

**Full Stack Development MERN- Lab Exercise****1. Write a program to insert image in HTML**

```
<!DOCTYPE html>

<html>

<body>

<h2>HTML Image</h2>



</body>

</html>
```

**2. Write a HTML program to develop a static Web Page for Catalog.**

```
<html>

<head>

<title>catalouge</title>

</head>

<body>

<center>

<table border=2 cellpadding=10 cellspacing=10>

<tr>

<td> </td>

<td>Book:XML Bible<br>

Author:Wingston<br>
```

Publication:wuely</td>

<td>\$40.5</td>

<td><input type="button" value="add to cart"></td>

</tr>

<tr>

<td> </td>

<td>Book:A1<br>

Author:S.Rusell<br>

Publication:Princeton hall</td>

<td>\$63</td>

<td><input type="button" value="add to cart"></td>

</tr>

<tr>

<td> </td>

<td>Book:JAVA 2<br>

Author:Watson<br>

Publication:BPB</td>

<td>\$35.5</td>

<td><input type="button" value="add to cart"></td>

</tr>

<tr>

<td> </td>

<td>Book:Html in 24 hrs<br>

Author:Sam Peter<br>

Publication:SAM</td>

<td>\$50</td>

<td><input type="button" value="add to cart"></td>

</tr>

</table>

</center>

</body>

</html>

### 3. Write a HTML program to develop a static Login Page.

<html>

<head>

<title>login</title>

</head>

<body>

<br><br><br><br>

<h1 align=center><u>LOGIN</u></h1>

<br><br><br>

<h4>

<center>

username&nbsp;&nbsp;&nbsp;<input type=text><br>

password&nbsp;&nbsp;&nbsp;<input type=password><br><br><br>

</h4>

<input type=submit value=submit>

&nbsp;&nbsp;&nbsp;

<input type=reset value=cancel>

</center>

```
</body>
</html>
```

#### 4. Write a HTML program to develop a static Registration Form.

```
<html>

<head>

<title>Registration</title>

</head>

<body bgcolor=lightblue>

<h1 align=center><u>Registration Form</u></h1>

<br><br><br>

<div>

<strong>
First Name &nbsp;<input type=text value=" " name="txt1"><br><br>
Last Name &nbsp;<input type=text value=" " name="txt2"><br><br>
UserName &nbsp;<input type=text value="" name="txt3"><br><br>
Password &nbsp;<input type=password value="" name="pwd1"><br>
Confirm Password &nbsp;<input type=password value="" name="pwd2"><br><br>
Address &nbsp;<textarea rows=3 cols=60></textarea><br><br>
Date of Birth &nbsp;<br>
dd<select name="sel1">

<option>--</option>

<option>01</option>

<option>02</option>
```

<option>03</option>

<option>04</option>

<option>05</option>

<option>27</option>

<option>28</option>

<option>29</option>

<option>30</option>

<option>31</option>

</select>

mm<select name="sel2">

<option>--</option>

<option>01</option>

<option>02</option>

<option>03</option>

<option>04</option>

<option>05</option>

<option>06</option>

<option>07</option>

<option>08</option>

<option>09</option>

<option>10</option>

<option>11</option>

<option>12</option>

</select>

yyyy<select name="sel3">

<option>-----</option>

<option>1987</option>

<option>1988</option>

<option>1989</option>

<option>1990</option>

<option>1991</option>

<option>1992</option>

<option>1993</option>

<option>1994</option>

<option>1995</option>

<option>1996</option>

<option>1997</option>

<option>1998</option>

<option>1999</option>

<option>2000</option>

<option>2001</option>

<option>2002</option>

<option>2003</option>

<option>2004</option>

<option>2005</option>

<option>2006</option>

<option>2007</option>

<option>2008</option>

<option>2009</option>

<option>2010</option>

<option>2011</option>

<option>2012</option>

<option>2013</option>

<option>2014</option>

<option>2015</option>

<option>2016</option>

<option>2017</option>

</select><br><br>

Sex &nbsp;

<input name="rb1" type="radio" value="radiobutton">Male

<input name="rb1" type="radio" value="radiobutton">Female

<br><br>

Martial Status &nbsp;

<input name="rb2" type="radio" value="radiobutton">Single

<input name="rb2" type="radio" value="radiobutton">Married

<br><br>

Mobile Number &nbsp; <input type="text" name="txt4"><br><br>

Branch &nbsp;

<input name="rb3" type="radio" value="radiobutton">CSE

```
<input name="rb3" type="radio" value="radiobutton">IT  
<input name="rb3" type="radio" value="radiobutton">ECE  
<input name="rb3" type="radio" value="radiobutton">EEE  
<input name="rb3" type="radio" value="radiobutton">MECH  
<br><br>
```

Languages Known &nbsp;nbsp;

```
<input name="cb1" type="checkbox" value="checkbox">English  
<input name="cb1" type="checkbox" value="checkbox">Telugu  
<input name="cb1" type="checkbox" value="checkbox">Hindi  
<input name="cb1" type="checkbox" value="checkbox">Kannada  
<input name="cb1" type="checkbox" value="checkbox">Tamil  
<br><br>  
<center>
```

```
<input type=submit value="SUBMIT" name="btn1">&nbsp;nbsp;  
<input type=reset value="CANCEL" name="btn1">  
  
</center>  
  
</strong>  
  
</body>  
  
</html>
```

## 5. Write a program to implement Inline, Internal and External

CSS.

CODE :

```
<html>

<head>

<title>CSS Example</title>

<!-- Internal CSS -->

<style>

/* Internal CSS */

h1 {

color: blue;

}

p {

font-size: 16px;

}

</style>

<!-- External CSS -->

<link rel="stylesheet" type="text/css" href="styles.css">

</head>

<body>

<!-- Inline CSS -->

<h1 style="font-family: Arial, sans-serif;">Inline CSS Example</h1>

<p style="color: green;">This is a paragraph with inline CSS.</p>

<!-- Internal CSS -->

<h1>Internal CSS Example</h1>

<p>This is a paragraph with internal CSS.</p>
```

```

<!-- External CSS -->

<h1 class="external">External CSS Example</h1>

<p class="external">This is a paragraph with external CSS.</p>

</body>

</html>

```

## JAVASCRIPT

### 6. Write a javascript program to validate USER LOGIN page.

```

<html>

<head>

<title>Login Validation</title>

<script language="javascript">
function formValidator()
{
    var username=document.getElementById('uname');
    var password=document.getElementById('pwd');
    if(isEmpty(username)&&isEmpty(password))
    {
        alert("enter something");
        document.form1.uname.focus();
    }
    if(!isEmpty(username)&&isEmpty(password)&&isAlphabet(username))
    {
        alert("Please enter password");
    }
}

```

```
document.form1.pwd.focus();

}

if(!isEmpty(username)&&!isEmpty(password)&&isAlphabet(username))

{

return true;

}

else

{

if(!isEmpty(username)&&!isEmpty(password)&&!isAlphabet(username))

{

alert("Please Enter only alphabets for username");

document.form1.uname.focus();

}

}

return false;

}

function isEmpty(elem)

{

if(elem.value.length==0)

{

return true;

}

return false;
```

```

    }

    function isAlphabet(elem)

    {

    var alphaExp=/^[a-z A-Z]+$/;

    if(elem.value.match(alphaExp))

    {

    return true;

    }

    }

    </script>

</head>

<body bgColor=megastar>

<h1 align=center>USER LOGIN VALIDATION</h1>

<br><br>

<form name="form1" onSubmit="return formValidator()">

<center>

<table border=0 colsSpacing=4>

<tr>

<td>Username:</td>

<td><input type=text value="" name="uname"></td>

</tr>

<tr>

<td>Password:</td>

```

```

<td><input type=password value="" name="pwd"></td>

</tr>

<tr>

<td><input type=submit value="SUBMIT" name="btn1"></td>

<td><input type=reset value="CANCEL" name="btn2"></td>

</tr>

</table>

</center>

</form>

</body>

</html>

```

## 7. Write a javascript program for validating REGISTRATION FORM.

```

<html>

<head>

<title>JavaScript sample registration form validation </title>

<script type='text/javascript'>

function formValidation()

{

var uid = document.form1.userid;

var passid = document.form1.passid;

var uname = document.form1.username;

var uadd = document.form1.address;

```

```
var uzip = document.form1.zip;

var uemail = document.form1.email;

var umsex = document.form1.msex;

var ufsex = document.form1.fsex;

if(userid_validation(uid,5,12))

{

if(userid_validation(passid,7,12))

{

if(allLetter(uname))

{

if(alphanumeric(uadd))

{

if(allNumeric(uzip))

{

if(ValidateEmail(uemail))

{

if(validsex(umsex,ufsex))

{

}

}

}

}

}

}

}
```

```

    }

    }

    return false;

    } function userid_validation(uid,mx,my)

    {

    var uid_len = uid.value.length;

    if (uid_len == 0 || uid_len >= my || uid_len < mx)

    {

    alert("It should not be empty / length be between "+mx+" to "+my);

    uid.focus();

    return false;

    }

    return true;

    }

    function allLetter(uname)

    {

    var letters = /^[A-Za-z]+$/;

    if(uname.value.match(letters))

    {

    return true;

    }

    else

    {

```

```

alert('Please input alphabet characters only');

uname.focus();

return false;

}

}

function alphanumeric(uadd)

{

var letters = /^[0-9a-zA-Z]+$/;

if(uadd.value.match(letters))

{

return true;

}

else

{

alert('Please input alphanumeric characters only');

uadd.focus();

return false;

}

}

function allnumeric(usize)

{

var numbers = /^[0-9]+$/;

if(usize.value.match(numbers))

```

```

{
return true;
}
else
{
alert('Please input numeric characters only');
uzip.focus();
return true;
}
}

```

```

function ValidateEmail(uemail)
{
var mailformat = /^[w+([\.-]?w+)*@w+([\.-]?w+)*(\.w{2,3})+$/;
if(uemail.value.match(mailformat))
{
return true;
}
else
{
alert("You have entered an invalid email address!");
uemail.focus();
return false;
}
}

```

```
} function validsex(umsex,ufsex)
```

```
{
```

```
x=0;
```

```
if(umsex.checked)
```

```
{
```

```
x++;
```

```
} if(ufsex.checked)
```

```
{
```

```
x++;
```

```
}
```

```
if(x==0)
```

```
{
```

```
alert('Select Male/Female');
```

```
umsex.focus();
```

```
return false;
```

```
}
```

```
else
```

```
{
```

```
return true;
```

```
}
```

```
}
```

```
</script>
```

```
</head>
```

```

<body>

<form name='form1' onsubmit='return formValidation()' >

<table width="500" cellpadding="3" style="border-collapse: collapse;">

<tr>

<td>User id </td>

<td><input type="text" name="userid" size="12" /></td>

</tr>

<tr>

<td>Password</td>

<td><input type="password" name="passid" size="12" /></td>

</tr>

<tr>

<td>Name</td>

<td><input type="text" name="username" size="50" /></td>

</tr>

<tr>

<td>Address</td>

<td><input type="text" name="address" size="50" /></td>

</tr>

<tr>

<td>ZIP Code </td>

<td><input type="text" name="zip" /></td>

</tr>

```

```

<tr>

<td>Email</td>

<td><input type="text" name="email" size="50" /></td>

</tr>

<tr>

<td>Sex</td>

<td><input type="radio" name="msex" value="Male" /> Male

<input type="radio" name="fsex" value="Female" /> Female</td>

</tr>

<tr>

<td>Language preference</td>

<td><input type="checkbox" name="en" value="en" checked />English

<input type="checkbox" name="nonen" value="noen" />Non English</td>

</tr>

<tr>

<td>Write about yourself<br>

(optional)</td>

<td><textarea name="desc" rows="4" cols="40"></textarea></td>

</tr>

<tr>

<td>&nbsp;</td>

<td><input type="submit" name="submit" value="Submit" /></td>

<td>&nbsp;</td>

```

```

</tr>

</table>

</form>

</body>

</html>

```

**8. Write a JavaScript program to swap the value of two variables without using**

a third variable. Take the value of both variables from the user.

```

<html>

<head>

<title>Javascript</title></head>

<body>

<script type="text/javascript">
var x=parseInt(prompt("Enter first number"));

var y=parseInt(prompt("Enter second number"));

x = x+y ;

y = x -y ;

x = x -y ;

document.write("x =");

document.write(x);

document.write("<br>");

2| 7-05-2023 / PREPARED BY: Tess Mary Thomas / ICT Dept.

document.write("y =");

document.write(y);

```

```
</script>
```

```
</body>
```

```
</html>
```

**9. Write a Javascript program to calculate the area of a square**

```
<html>
```

```
<head>
```

```
<title>Area of square</title></head>
```

```
<body>
```

```
<script type="text/javascript">
```

```
var s=parseInt(prompt("enter the side of square"));
```

```
var area;
```

```
area=s*s;
```

```
document.write("Area of square=");
```

```
document.write(area);
```

```
</script>
```

```
</body></html>
```

**Write a Javascript program to convert Kilometers to centimetrs**

```
<html>
```

```
<head>
```

```
<title>javascript</title></head>
```

```
<body>
```

```
<script type="text/javascript">
```

```
var km=parseInt(prompt("enter the value in Kilometer"));
```

```
var cm;  
  
cm=100000*km;  
  
document.write("kilometre to centimetre=");  
  
document.write(cm);  
  
</script>  
  
</body>  
  
</html>
```

**10. Develop a JavaScript program to perform following arithmetic operations:**

addition, subtraction, multiplication and division.

```
<html>  
  
<head>  
<title>Arithmetic operators in Javascript</title></head>  
  
<body>  
  
<script type="text/javascript">  
  
var a=parseInt(prompt("enter first number"));  
  
var b=parseInt(prompt("enter second number"));  
  
var sum;  
  
var difference;  
  
var product;  
  
var quotient;  
  
sum=a+b;  
  
difference=a-b;  
  
product=a*b;
```

```
Quotient=a/b;

document.write("Sum= ");

document.write(sum);

document.write("<br>");

document.write("Difference =");

document.write(difference);

document.write("<br>");

document.write("Product=");

document.write(product);

document.write("<br>");

document.write("Quotient=");

document.write(quotient);

</script>

</body>

</html>
```

## REACT

**11. Develop React JS program that allows user to input their first name and last name and display their full name. Note: The default values for first name and last n**

### Step 1: Set Up Your React App

If you haven't already created a React app, you can do so with Create React App. Open your terminal and run:

```
npx create-react-app name-display-app
```

```
cd name-display-app
```

## Step 2: Modify the App.js

Replace the content of src/App.js with the following code:

```
import React, { useState } from 'react';
```

```
function App() {
```

```
  // Default values for first name and last name
```

```
  const [firstName, setFirstName] = useState('John');
```

```
  const [lastName, setLastName] = useState('Doe');
```

```
  // Function to handle input change
```

```
  const handleFirstNameChange = (e) => {
```

```
    setFirstName(e.target.value);
```

```
  };
```

```
  const handleLastNameChange = (e) => {
```

```
    setLastName(e.target.value);
```

```
  };
```

```
// Full name

const fullName = `${firstName} ${lastName}`;

return (

  <div style={{ padding: '20px' }}>

    <h1>Full Name Display</h1>

    <div>

      <label>

        First Name:

        <input

          type="text"

          value={firstName}

          onChange={handleFirstNameChange}

        />

      </label>

    </div>

    <div>

      <label>

        Last Name:

        <input

          type="text"

          value={lastName}

          onChange={handleLastNameChange}
```

```

    />

    </label>

  </div>

  <h2>Your Full Name: {fullName}</h2>

</div>

);
}

export default App;

```

### Step 3: Run the App

In your terminal, run the following command to start the development server:

```
npm start
```

**12. Develop an React JS application that displays a list of shopping items. Allow users to add and remove items from the list using directives and controllers. Note: The default values of items may be included in the program.**

### Step 1: Set Up Your React App

If you haven't already created a React app, you can do so with Create React App. Open your terminal and run:

```
npm create-react-app shopping-list-app
```

```
cd shopping-list-app
```

### Step 2: Modify the App.js

Replace the content of src/App.js with the following code:

```

import React, { useState } from 'react';

function App() {

  // Default shopping items

  const [items, setItems] = useState(['Apples', 'Bananas', 'Carrots']);

  const [newItem, setNewItem] = useState("");

  // Function to add an item

  const addItem = () => {

    if (newItem.trim()) {

      setItems([...items, newItem.trim()]);

      setNewItem("");

    }

  };

  // Function to remove an item

  const removeItem = (index) => {

    const updatedItems = items.filter((_, i) => i !== index);

    setItems(updatedItems);

  };

  return (

    <div style={{ padding: '20px' }}>

      <h1>Shopping List</h1>

```

```

<div>

  <input

    type="text"

    value={newItem}

    onChange={(e) => setNewItem(e.target.value)}

    placeholder="Add new item"

  />

  <button onClick={addItem}>Add Item</button>

</div>

<ul style={{ listStyleType: 'none', padding: 0 }}>

  {items.map((item, index) => (

    <li key={index} style={{ display: 'flex', justifyContent: 'space-between' }}>

      {item}

      <button onClick={() => removeItem(index)}>Remove</button>

    </li>

  ))}

</ul>

</div>

);

}

export default App;

```

### Step 3: Run the App

In your terminal, run the following command to start the development server:

```
npm start
```

**13. Develop a simple React JS calculator application that can perform basic mathematical operations (addition, subtraction, multiplication, division) based on user input.**

**Step 1: Set Up Your React App**

If you haven't already created a React app, you can do so with Create React App. Open your terminal and run:

```
npx create-react-app calculator-app
```

```
cd calculator-app
```

**Step 2: Modify the App.js**

Replace the content of src/App.js with the following code:

```
import React, { useState } from 'react';

function App() {

  const [num1, setNum1] = useState("");

  const [num2, setNum2] = useState("");

  const [result, setResult] = useState(null);

  const [operation, setOperation] = useState("");

  const calculate = () => {

    const number1 = parseFloat(num1);

    const number2 = parseFloat(num2);
```

```
if (isNaN(number1) || isNaN(number2)) {  
    setResult('Please enter valid numbers');  
    return;  
}
```

```
let res;
```

```
switch (operation) {
```

```
    case '+':
```

```
        res = number1 + number2;
```

```
        break;
```

```
    case '-':
```

```
        res = number1 - number2;
```

```
        break;
```

```
    case '*':
```

```
        res = number1 * number2;
```

```
        break;
```

```
    case '/':
```

```
        if (number2 === 0) {
```

```
            setResult('Cannot divide by zero');
```

```
            return;
```

```
        }
```

```
        res = number1 / number2;
```

```

    break;

    default:

        setResult('Select an operation');

        return;

    }

```

```

    setResult(res);

};

```

```

return (

```

```

    <div style={{ padding: '20px' }}>

```

```

        <h1>Simple Calculator</h1>

```

```

        <div>

```

```

            <input

```

```

                type="number"

```

```

                value={num1}

```

```

                onChange={(e) => setNum1(e.target.value)}

```

```

                placeholder="First Number"

```

```

            />

```

```

        </div>

```

```

    <div>

```

```

        <input

```

```

            type="number"

```

```

    value={num2}

    onChange={(e) => setNum2(e.target.value)}

    placeholder="Second Number"

  />

</div>

<div>

  <select onChange={(e) => setOperation(e.target.value)} value={operation}>

    <option value="">Select Operation</option>

    <option value="+">+</option>

    <option value="-">-</option>

    <option value="*">*</option>

    <option value="/">/</option>

  </select>

</div>

<button onClick={calculate}>Calculate</button>

{result !== null && <h2>Result: {result}</h2>}

</div>

);

}

```

```
export default App;
```

### Step 3: Run the App

In your terminal, run the following command to start the development server:

npm start

#### 14. Develop React JS application that displays details of students and their CGPA.

**Allow users to read the number of students and display the count. Note: Student details may be included in the program.**

##### Step 1: Set Up Your React App

If you haven't already created a React app, you can do so with Create React App. Open your terminal and run:

```
npx create-react-app student-cgpa-app
```

```
cd student-cgpa-app
```

##### Step 2: Modify the App.js

Replace the content of src/App.js with the following code:

```
import React from 'react';

function App() {

  // Sample student data

  const students = [

    { id: 1, name: 'Alice Johnson', cgpa: 3.8 },

    { id: 2, name: 'Bob Smith', cgpa: 3.6 },

    { id: 3, name: 'Charlie Brown', cgpa: 3.9 },

    { id: 4, name: 'David Wilson', cgpa: 3.5 },

    { id: 5, name: 'Eve Davis', cgpa: 3.7 },

  ];
```

```
// Count of students

const studentCount = students.length;

return (

  <div style={{ padding: '20px' }}>

    <h1>Student CGPA Details</h1>

    <h2>Total Students: {studentCount}</h2>

    <ul style={{ listStyleType: 'none', padding: 0 }}>

      {students.map(student => (

        <li key={student.id} style={{ margin: '10px 0' }}>

          <strong>{student.name}</strong> - CGPA: {student.cgpa}

        </li>

      ))}

    </ul>

  </div>

);

}
```

export default App;

### Step 3: Run the App

In your terminal, run the following command to start the development server:

```
npm start
```

## MongoDB

- 15. Write a MongoDB query to display all the documents in the collection restaurants.Best restaurants near me**

Structure of 'restaurants' collection:

```
{
  "address": {
    "building": "1007",
    "coord": [ -73.856077, 40.848447 ],
    "street": "Morris Park Ave",
    "zipcode": "10462"
  },
  "borough": "Bronx",
  "cuisine": "Bakery",
  "grades": [
    { "date": { "$date": 1393804800000 }, "grade": "A", "score": 2 },
    { "date": { "$date": 1378857600000 }, "grade": "A", "score": 6 },
    { "date": { "$date": 1358985600000 }, "grade": "A", "score": 10 },
    { "date": { "$date": 1322006400000 }, "grade": "A", "score": 9 },
    { "date": { "$date": 1299715200000 }, "grade": "B", "score": 14 }
  ],
  "name": "Morris Park Bake Shop",
```

```
"restaurant_id": "30075445"
```

```
}
```

Query:

```
db.restaurants.find();
```

Output:

```
{ "_id" : ObjectId("564c2d939eb21ad392f175c9"), "address" : { "building" : "351",
"coord" : [ -73.98513559999999, 40.7676919 ], "street" : "West 57 Street", "zipcode" :
"10019" }, "borough" : "Manhattan", "cuisine" : "Irish", "grades" : [ { "date" :
ISODate("2014-09-06T00:00:00Z"), "grade" : "A", "score" : 2 }, { "date" :
ISODate("2013-07-22T00:00:00Z"), "grade" : "A", "score" : 11 }, { "date" :
ISODate("2012-07-31T00:00:00Z"), "grade" : "A", "score" : 12 }, { "date" :
ISODate("2011-12-29T00:00:00Z"), "grade" : "A", "score" : 12 } ], "name" : "Dj Reynolds Pub And Restaurant",
"restaurant_id" : "30191841" }
{ "_id" : ObjectId("564c2d939eb21ad392f175ca"), "address" : { "building" : "1007",
"coord" : [ -73.856077, 40.848447 ], "street" : "Morris Park Ave", "zipcode" : "10462" },
"borough" : "Bronx", "cuisine" : "Bakery", "grades" : [ { "date" : ISODate("2014-03-
03T00:00:00Z"), "grade" : "A", "score" : 2 }, { "date" : ISODate("2013-09-
11T00:00:00Z"), "grade" : "A", "score" : 6 }, { "date" : ISODate("2013-01-
24T00:00:00Z"), "grade" : "A", "score" : 10 }, { "date" : ISODate("2011-11-
23T00:00:00Z"), "grade" : "A", "score" : 9 }, { "date" : ISODate("2011-03-10T00:00:00Z"), "grade" : "B",
"score" : 14 } ], "name" : "Morris Park Bake Shop", "restaurant_id" : "30075445" }
{ "_id" : ObjectId("564c2d939eb21ad392f175cb"), "address" : { "building" : "2780",
"coord" : [ -73.98241999999999, 40.579505 ], "street" : "Stillwell Avenue", "zipcode" :
"11224" }, "borough" : "Brooklyn", "cuisine" : "American ", "grades" : [ { "date" :
ISODate("2014-06-10T00:00:00Z"), "grade" : "A", "score" : 5 }, { "date" :
ISODate("2013-06-05T00:00:00Z"), "grade" : "A", "score" : 7 }, { "date" :
```

```

ISODate("2012-04-13T00:00:00Z"), "grade" : "A", "score" : 12 }, { "date" :
ISODate("2011-10-12T00
:00:00Z"), "grade" : "A", "score" : 12 } ], "name" : "Riviera Caterer", "restaurant_id" :
"40356018" }

{ "_id" : ObjectId("564c2d939eb21ad392f175cc"), "address" : { "building" : "469",
"coord" : [ -73.961704, 40.662942 ], "street" : "Flatbush Avenue", "zipcode" : "11225" },
"borough" : "Brooklyn", "cuisine" : "Hamburgers", "grades" : [ { "date" :
ISODate("2014-12-30T00:00:00Z"), "grade" : "A", "score" : 8 }, { "date" :
ISODate("2014-07-01T00:00:00Z"), "grade" : "B", "score" : 23 }, { "date" :
ISODate("2013-04-30T00:00:00Z"), "grade" : "A", "score" : 12 }, { "date" :
ISODate("2012-05-08T00:00:00Z"
), "grade" : "A", "score" : 12 } ], "name" : "Wendy'S", "restaurant_id" : "30112340" }

{ "_id" : ObjectId("564c2d939eb21ad392f175cd"), "address" : { "building" : "97-22",
"coord" : [ -73.8601152, 40.7311739 ], "street" : "63 Road", "zipcode" : "11374" },
"borough" : "Queens", "cuisine" : "Jewish/Kosher", "grades" : [ { "date" :
ISODate("2014-11-24T00:00:00Z"), "grade" : "Z", "score" : 20 }, { "date" :
ISODate("2013-01-17T00:00:00Z"), "grade" : "A", "score" : 13 }, { "date" :
ISODate("2012-08-02T00:00:00Z"), "grade" : "A", "score" : 13 }, { "date" :
ISODate("2011-12-15T00:00:00Z"),
"grade" : "B", "score" : 25 } ], "name" : "Tov Kosher Kitchen", "restaurant_id" :
"40356068" }

{ "_id" : ObjectId("564c2d939eb21ad392f175ce"), "address" : { "building" : "8825",
"coord" : [ -73.8803827, 40.7643124 ], "street" : "Astoria Boulevard", "zipcode" :
"11369" }, "borough" : "Queens", "cuisine" : "American ", "grades" : [ { "date" :
ISODate("2014-11-15T00:00:00Z"), "grade" : "Z", "score" : 38 }, { "date" :
ISODate("2014-05-02T00:00:00Z"), "grade" : "A", "score" : 10 }, { "date" :
ISODate("2013-03-02T00:00:00Z"), "grade" : "A", "score" : 7 }, { "date" :
ISODate("2012-02-10T00:00:00

```

```

Z"), "grade" : "A", "score" : 13 } ], "name" : "Brunos On The Boulevard",
"restaurant_id" : "40356151" }

{ "_id" : ObjectId("564c2d939eb21ad392f175cf"), "address" : { "building" : "6409",
"coord" : [ -74.00528899999999, 40.628886 ], "street" : "11 Avenue", "zipcode" :
"11219" }, "borough" : "Brooklyn", "cuisine" : "American ", "grades" : [ { "date" :
ISODate("2014-07-18T00:00:00Z"), "grade" : "A", "score" : 12 }, { "date" :
ISODate("2013-07-30T00:00:00Z"), "grade" : "A", "score" : 12 }, { "date" :
ISODate("2013-02-13T00:00:00Z"), "grade" : "A", "score" : 11 }, { "date" :
ISODate("2012-08-16T00:00:0
0Z"), "grade" : "A", "score" : 2 }, { "date" : ISODate("2011-08-17T00:00:00Z"),
"grade" : "A", "score" : 11 } ], "name" : "Regina Caterers", "restaurant_id" :
"40356649" }

{ "_id" : ObjectId("564c2d939eb21ad392f175d0"), "address" : { "building" : "7114",
"coord" : [ -73.9068506, 40.6199034 ], "street" : "Avenue U", "zipcode" : "11234" },
"borough" : "Brooklyn", "cuisine" : "Delicatessen", "grades" : [ { "date" :
ISODate("2014-05-29T00:00:00Z"), "grade" : "A", "score" : 10 }, { "date" :
ISODate("2014-01-14T00:00:00Z"), "grade" : "A", "score" : 10 }, { "date" :
ISODate("2013-08-03T00:00:00Z"), "grade" : "A", "score" : 8 }, { "date" :
ISODate("2012-07-18T00:00:00Z"),

"grade" : "A", "score" : 10 }, { "date" : ISODate("2012-03-09T00:00:00Z"), "grade" :
"A", "score" : 13 }, { "date" : ISODate("2011-10-14T00:00:00Z"), "grade" : "A",
"score" : 9 } ], "name" : "Wilken'S Fine Food", "restaurant_id" : "40356483" }

{ "_id" : ObjectId("564c2d939eb21ad392f175d1"), "address" : { "building" : "2300",
"coord" : [ -73.8786113, 40.8502883 ], "street" : "Southern Boulevard", "zipcode" :
"10460" }, "borough" : "Bronx", "cuisine" : "American ", "grades" : [ { "date" :
ISODate("2014-05-28T00:00:00Z"), "grade" : "A", "score" : 11 }, { "date" :
ISODate("2013-06-19T00:00:00Z"), "grade" : "A", "score" : 4 }, { "date" :
ISODate("2012-06-15T00:00:00Z"), "grade" : "A", "score" : 3 } ], "name" : "Wild Asia",
"restaurant_id"

```

```

: "40357217" }

{ "_id" : ObjectId("564c2d939eb21ad392f175d2"), "address" : { "building" : "1839",
"coord" : [ -73.9482609, 40.6408271 ], "street" : "Nostrand Avenue", "zipcode" :
"11226" }, "borough" : "Brooklyn", "cuisine" : "Ice Cream, Gelato, Yogurt, Ices",
"grades" : [ { "date" : ISODate("2014-07-14T00:00:00Z"), "grade" : "A", "score" : 12 },
{ "date" : ISODate("2013-07-10T00:00:00Z"), "grade" : "A", "score" : 8 }, { "date" :
ISODate("2012-07-11T00:00:00Z"), "grade" : "A", "score" : 5 }, { "date" : ISODate
("2012-02-23T00:00:00Z"), "grade" : "A", "score" : 8 } ], "name" : "Taste The Tropics
Ice Cream", "restaurant_id" : "40356731" }

{ "_id" : ObjectId("564c2d939eb21ad392f175d3"), "address" : { "building" : "7715",
"coord" : [ -73.9973325, 40.611748899999999 ], "street" : "18 Avenue", "zipcode" :
"11214" }, "borough" : "Brooklyn", "cuisine" : "American ", "grades" : [ { "date" :
ISODate("2014-04-16T00:00:00Z"), "grade" : "A", "score" : 5 }, { "date" :
ISODate("2013-04-23T00:00:00Z"), "grade" : "A", "score" : 2 }, { "date" :
ISODate("2012-04-24T00:00:00Z"), "grade" : "A", "score" : 5 }, { "date" :
ISODate("2011-12-16T00:00:00Z
"), "grade" : "A", "score" : 2 } ], "name" : "C & C Catering Service", "restaurant_id" :
"40357437" }

{ "_id" : ObjectId("564c2d939eb21ad392f175d4"), "address" : { "building" : "1269",
"coord" : [ -73.871194, 40.6730975 ], "street" : "Sutter Avenue", "zipcode" : "11208" },
"borough" : "Brooklyn", "cuisine" : "Chinese", "grades" : [ { "date" : ISODate("2014-
09-16T00:00:00Z"), "grade" : "B", "score" : 21 }, { "date" : ISODate("2013-08-
28T00:00:00Z"), "grade" : "A", "score" : 7 }, { "date" : ISODate("2013-04-
02T00:00:00Z"), "grade" : "C", "score" : 56 }, { "date" : ISODate("2012-08-
15T00:00:00Z"),
"grade" : "B", "score" : 27 }, { "date" : ISODate("2012-03-28T00:00:00Z"), "grade" :
"B", "score" : 27 } ], "name" : "May May Kitchen", "restaurant_id" : "40358429" }

```

```
{ "_id" : ObjectId("564c2d939eb21ad392f175d5"), "address" : { "building" : "1",
"coord" : [ -73.96926909999999, 40.7685235 ], "street" : "East 66 Street", "zipcode" :
"10065" }, "borough" : "Manhattan", "cuisine" : "American ", "grades" : [ { "date" :
ISODate("2014-05-07T00:00:00Z"), "grade" : "A", "score" : 3 }, { "date" :
ISODate("2013-05-03T00:00:00Z"), "grade" : "A", "score" : 4 }, { "date" :
ISODate("2012-04-30T00:00:00Z"), "grade" : "A", "score" : 6 }, { "date" :
ISODate("2011-12-27T00:0
0:00Z"), "grade" : "A", "score" : 0 } ], "name" : "1 East 66Th Street Kitchen",
"restaurant_id" : "40359480" }

{ "_id" : ObjectId("564c2d939eb21ad392f175d6"), "address" : { "building" : "705",
"coord" : [ -73.9653967, 40.6064339 ], "street" : "Kings Highway", "zipcode" :
"11223" }, "borough" : "Brooklyn", "cuisine" : "Jewish/Kosher", "grades" : [ { "date" :
ISODate("2014-11-10T00:00:00Z"), "grade" : "A", "score" : 11 }, { "date" :
ISODate("2013-10-10T00:00:00Z"), "grade" : "A", "score" : 13 }, { "date" :
ISODate("2012-10-04T00:00:00Z"), "grade" : "A", "score" : 7 }, { "date" :
ISODate("2012-05-21T00:00:0
0Z"), "grade" : "A", "score" : 9 }, { "date" : ISODate("2011-12-30T00:00:00Z"),
"grade" : "B", "score" : 19 } ], "name" : "Seuda Foods", "restaurant_id" : "40360045" }

{ "_id" : ObjectId("564c2d949eb21ad392f1c593"), "address" : { "building" : "42",
"coord" : [ -73.82948700000001, 40.657432 ], "street" : "Broadway", "zipcode" :
"11414" }, "borough" : "Queens", "cuisine" : "Other", "grades" : [ ], "name" : "Laquana
King", "restaurant_id" : "50003441" }

{ "_id" : ObjectId("564c2d939eb21ad392f175d7"), "address" : { "building" : "203",
"coord" : [ -73.97822040000001, 40.6435254 ], "street" : "Church Avenue", "zipcode" :
"11218" }, "borough" : "Brooklyn", "cuisine" : "Ice Cream, Gelato, Yogurt, Ices",
"grades" : [ { "date" : ISODate("2014-02-10T00:00:00Z"), "grade" : "A", "score" : 2 },
{ "date" : ISODate("2013-01-02T00:00:00Z"), "grade" : "A", "score" : 13 }, { "date" :
ISODate("2012-01-09T00:00:00Z"), "grade" : "A", "score" : 3 }, { "date" : ISO
```

```

Date("2011-11-07T00:00:00Z"), "grade" : "P", "score" : 12 }, { "date" : ISODate("2011-
07-21T00:00:00Z"), "grade" : "A", "score" : 13 } ], "name" : "Carvel Ice Cream",
"restaurant_id" : "40360076" }

{ "_id" : ObjectId("564c2d939eb21ad392f175d8"), "address" : { "building" : "265-15",
"coord" : [ -73.7032601, 40.7386417 ], "street" : "Hillside Avenue", "zipcode" :
"11004" }, "borough" : "Queens", "cuisine" : "Ice Cream, Gelato, Yogurt, Ices",
"grades" : [ { "date" : ISODate("2014-10-28T00:00:00Z"), "grade" : "A", "score" : 9 },
{ "date" : ISODate("2013-09-18T00:00:00Z"), "grade" : "A", "score" : 10 }, { "date" :
ISODate("2012-09-20T00:00:00Z"), "grade" : "A", "score" : 13 } ], "name" : "Carve
l Ice Cream", "restaurant_id" : "40361322" }

{ "_id" : ObjectId("564c2d939eb21ad392f175d9"), "address" : { "building" : "6909",
"coord" : [ -74.0259567, 40.6353674 ], "street" : "3 Avenue", "zipcode" : "11209" },
"borough" : "Brooklyn", "cuisine" : "Delicatessen", "grades" : [ { "date" :
ISODate("2014-08-21T00:00:00Z"), "grade" : "A", "score" : 4 }, { "date" :
ISODate("2014-03-05T00:00:00Z"), "grade" : "A", "score" : 3 }, { "date" :
ISODate("2013-01-10T00:00:00Z"), "grade" : "A", "score" : 10 } ], "name" : "Nordic
Delicacies", "restaurant_
id" : "40361390" }

{ "_id" : ObjectId("564c2d939eb21ad392f175da"), "address" : { "building" : "284",
"coord" : [ -73.9829239, 40.6580753 ], "street" : "Prospect Park West", "zipcode" :
"11215" }, "borough" : "Brooklyn", "cuisine" : "American ", "grades" : [ { "date" :
ISODate("2014-11-19T00:00:00Z"), "grade" : "A", "score" : 11 }, { "date" :
ISODate("2013-11-14T00:00:00Z"), "grade" : "A", "score" : 2 }, { "date" :
ISODate("2012-12-05T00:00:00Z"), "grade" : "A", "score" : 13 }, { "date" :
ISODate("2012-05-17T00:00:
00Z"), "grade" : "A", "score" : 11 } ], "name" : "The Movable Feast", "restaurant_id" :
"40361606" }

```

```
{ "_id" : ObjectId("564c2d939eb21ad392f175db"), "address" : { "building" : "522",
"coord" : [ -73.95171, 40.767461 ], "street" : "East 74 Street", "zipcode" : "10021" },
"borough" : "Manhattan", "cuisine" : "American ", "grades" : [ { "date" :
ISODate("2014-09-02T00:00:00Z"), "grade" : "A", "score" : 12 }, { "date" :
ISODate("2013-12-19T00:00:00Z"), "grade" : "B", "score" : 16 }, { "date" :
ISODate("2013-05-28T00:00:00Z"), "grade" : "A", "score" : 9 }, { "date" :
ISODate("2012-12-07T00:00:00Z"
), "grade" : "A", "score" : 13 }, { "date" : ISODate("2012-03-29T00:00:00Z"), "grade" :
"A", "score" : 11 } ], "name" : "Glorious Food", "restaurant_id" : "40361521"
```

16. Write a MongoDB query to display the fields restaurant\_id, name, borough and cuisine for all the documents in the collection restaurant.Best restaurants near me

Structure of 'restaurants' collection :

```
{
  "address": {
    "building": "1007",
    "coord": [ -73.856077, 40.848447 ],
    "street": "Morris Park Ave",
    "zipcode": "10462"
  },
  "borough": "Bronx",
  "cuisine": "Bakery",
  "grades": [
    { "date": { "$date": 1393804800000 }, "grade": "A", "score": 2 },
```

```
{ "date": { "$date": 1378857600000 }, "grade": "A", "score": 6 },
{ "date": { "$date": 1358985600000 }, "grade": "A", "score": 10 },
{ "date": { "$date": 1322006400000 }, "grade": "A", "score": 9 },
{ "date": { "$date": 1299715200000 }, "grade": "B", "score": 14 }
],
"name": "Morris Park Bake Shop",
"restaurant_id": "30075445"
}
```

Query:

```
db.restaurants.find({}, {"restaurant_id" : 1, "name": 1, "borough": 1, "cuisine" : 1});
```

Output:

```
{ "_id" : ObjectId("564c2d939eb21ad392f175c9"), "borough" : "Manhattan", "cuisine" : "Irish",
"name" : "Dj Reynolds Pub And Restaurant", "restaurant_id" : "30191841" }

{ "_id" : ObjectId("564c2d939eb21ad392f175ca"), "borough" : "Bronx", "cuisine" : "Bakery",
"name" : "Morris Park Bake Shop", "restaurant_id" : "30075445" }

{ "_id" : ObjectId("564c2d939eb21ad392f175cb"), "borough" : "Brooklyn", "cuisine" :
"American ", "name" : "Riviera Caterer", "restaurant_id" : "40356018" }

{ "_id" : ObjectId("564c2d939eb21ad392f175cc"), "borough" : "Brooklyn", "cuisine" :
"Hamburgers", "name" : "Wendy'S", "restaurant_id" : "30112340" }

{ "_id" : ObjectId("564c2d939eb21ad392f175cd"), "borough" : "Queens", "cuisine" :
"Jewish/Kosher", "name" : "Tov Kosher Kitchen", "restaurant_id" : "40356068" }

{ "_id" : ObjectId("564c2d939eb21ad392f175ce"), "borough" : "Queens", "cuisine" : "American
", "name" : "Brunos On The Boulevard", "restaurant_id" : "40356151" }
```

```
{ "_id" : ObjectId("564c2d939eb21ad392f175cf"), "borough" : "Brooklyn", "cuisine" :
"American ", "name" : "Regina Caterers", "restaurant_id" : "40356649" }

{ "_id" : ObjectId("564c2d939eb21ad392f175d0"), "borough" : "Brooklyn", "cuisine" :
"Delicatessen", "name" : "Wilken'S Fine Food", "restaurant_id" : "40356483" }

{ "_id" : ObjectId("564c2d939eb21ad392f175d1"), "borough" : "Bronx", "cuisine" : "American ",
"name" : "Wild Asia", "restaurant_id" : "40357217" }

{ "_id" : ObjectId("564c2d939eb21ad392f175d2"), "borough" : "Brooklyn", "cuisine" : "Ice
Cream, Gelato, Yogurt, Ices", "name" : "Taste The Tropics Ice Cream", "restaurant_id" :
"40356731" }

{ "_id" : ObjectId("564c2d939eb21ad392f175d3"), "borough" : "Brooklyn", "cuisine" :
"American ", "name" : "C & C Catering Service", "restaurant_id" : "40357437" }

{ "_id" : ObjectId("564c2d939eb21ad392f175d4"), "borough" : "Brooklyn", "cuisine" :
"Chinese", "name" : "May May Kitchen", "restaurant_id" : "40358429" }

{ "_id" : ObjectId("564c2d939eb21ad392f175d5"), "borough" : "Manhattan", "cuisine" :
"American ", "name" : "1 East 66Th Street Kitchen", "restaurant_id" : "40359480" }

{ "_id" : ObjectId("564c2d939eb21ad392f175d6"), "borough" : "Brooklyn", "cuisine" :
"Jewish/Kosher", "name" : "Seuda Foods", "restaurant_id" : "40360045" }

{ "_id" : ObjectId("564c2d949eb21ad392f1c593"), "borough" : "Queens", "cuisine" : "Other",
"name" : "Laquana King", "restaurant_id" : "50003441" }

{ "_id" : ObjectId("564c2d939eb21ad392f175d7"), "borough" : "Brooklyn", "cuisine" : "Ice
Cream, Gelato, Yogurt, Ices", "name" : "Carvel Ice Cream", "restaurant_id" : "40360076" }

{ "_id" : ObjectId("564c2d939eb21ad392f175d8"), "borough" : "Queens", "cuisine" : "Ice Cream,
Gelato, Yogurt, Ices", "name" : "Carvel Ice Cream", "restaurant_id" : "40361322" }

{ "_id" : ObjectId("564c2d939eb21ad392f175d9"), "borough" : "Brooklyn", "cuisine" :
"Delicatessen", "name" : "Nordic Delicacies", "restaurant_id" : "40361390" }
```

```
{ "_id" : ObjectId("564c2d939eb21ad392f175da"), "borough" : "Brooklyn", "cuisine" :
"American ", "name" : "The Movable Feast", "restaurant_id" : "40361606" }

{ "_id" : ObjectId("564c2d939eb21ad392f175db"), "borough" : "Manhattan", "cuisine" :
"American ", "name" : "Glorious Food", "restaurant_id" : "40361521" }
```

17. Write a MongoDB query to find the restaurants that do not prepare any cuisine of 'American' and their grade score more than 70 and latitude less than -65.754168.

Structure of 'restaurants' collection :

```
{
  "address": {
    "building": "1007",
    "coord": [ -73.856077, 40.848447 ],
    "street": "Morris Park Ave",
    "zipcode": "10462"
  },
  "borough": "Bronx",
  "cuisine": "Bakery",
  "grades": [
    { "date": { "$date": 1393804800000 }, "grade": "A", "score": 2 },
    { "date": { "$date": 1378857600000 }, "grade": "A", "score": 6 },
    { "date": { "$date": 1358985600000 }, "grade": "A", "score": 10 },
    { "date": { "$date": 1322006400000 }, "grade": "A", "score": 9 },
    { "date": { "$date": 1299715200000 }, "grade": "B", "score": 14 }
  ],
  "name": "Morris Park Bake Shop",
  "restaurant_id": "30075445"
```

```
}
```

Query:

```
db.restaurants.find(
  {$and:
    [
      {"cuisine" : {$ne : "American "}},
      {"grades.score" : {$gt : 70}},
      {"address.coord" : {$lt : -65.754168}}
    ]
  }
);
```

Output:

```
{ "_id" : ObjectId("564c2d939eb21ad392f177c8"), "address" : { "building" : "345",
"coord" : [ -73.9864626, 40.7266739 ], "street" : "East 6 Street", "zipcode" :
"10003" }, "borough" : "Manhattan", "cuisine" : "Indian", "grades" : [ { "date" :
ISODate("2014-09-15T00:00:00Z"), "grade" : "A", "score" : 5 }, { "date" :
ISODate("2014-01-14T00:00:00Z"), "grade" : "A", "score" : 8 }, { "date" :
ISODate("2013-05-30T00:00:00Z"), "grade" : "A", "score" : 12 }, { "date" :
ISODate("2013-04-24T00:00:00Z"), "
grade" : "P", "score" : 2 }, { "date" : ISODate("2012-10-01T00:00:00Z"), "grade" :
"A", "score" : 9 }, { "date" : ISODate("2012-04-06T00:00:00Z"), "grade" : "C",
"score" : 92 }, { "date" : ISODate("2011-11-03T00:00:00Z"), "grade" : "C", "score" :
41 } ], "name" : "Gandhi", "restaurant_id" : "40381295" }
{ "_id" : ObjectId("564c2d939eb21ad392f17929"), "address" : { "building" : "130",
"coord" : [ -73.984758, 40.7457939 ], "street" : "Madison Avenue", "zipcode" :
"10016" }, "borough" : "Manhattan", "cuisine" : "Pizza/Italian", "grades" : [ { "date" :
ISODate("2014-12-24T00:00:00Z"), "grade" : "Z", "score" : 31 }, { "date" :
```

```

ISODate("2014-06-17T00:00:00Z"), "grade" : "C", "score" : 98 }, { "date" :
ISODate("2013-12-12T00:00:00Z"), "grade" : "C", "score" : 32 }, { "date" :
ISODate("2013-05-22T00:00
:00Z"), "grade" : "B", "score" : 21 }, { "date" : ISODate("2012-05-02T00:00:00Z"),
"grade" : "A", "score" : 11 } ], "name" : "Bella Napoli", "restaurant_id" :
"40393488" }
{ "_id" : ObjectId("564c2d939eb21ad392f17935"), "address" : { "building" : "101",
"coord" : [ -73.9243061, 40.8276297 ], "street" : "East 161 Street", "zipcode" :
"10451" }, "borough" : "Bronx", "cuisine" : "Latin (Cuban, Dominican, Puerto Rican,
South & Central American)", "grades" : [ { "date" : ISODate("2014-04-
10T00:00:00Z"), "grade" : "A", "score" : 10 }, { "date" : ISODate("2013-10-
01T00:00:00Z"), "grade" : "A", "score" : 6 }, { "date" : ISODate("2013-04-
11T00:00:00Z"), "grade" : "B", "sco
re" : 25 }, { "date" : ISODate("2012-10-25T00:00:00Z"), "grade" : "A", "score" :
12 }, { "date" : ISODate("2012-10-10T00:00:00Z"), "grade" : "P", "score" : 12 },
{ "date" : ISODate("2012-05-25T00:00:00Z"), "grade" : "B", "score" : 14 }, { "date" :
ISODate("2011-09-14T00:00:00Z"), "grade" : "B", "score" : 26 }, { "date" :
ISODate("2011-04-25T00:00:00Z"), "grade" : "C", "score" : 76 } ], "name" : "El
Molino Rojo Restaurant", "restaurant_id" : "40393688" }
{ "_id" : ObjectId("564c2d949eb21ad392f17a86"), "address" : { "building" : "289",
"coord" : [ -73.94610279999999, 40.7137587 ], "street" : "Manhattan Avenue",
"zipcode" : "11211" }, "borough" : "Brooklyn", "cuisine" : "Bakery", "grades" :
[ { "date" : ISODate("2014-03-19T00:00:00Z"), "grade" : "A", "score" : 10 },
{ "date" : ISODate("2013-10-10T00:00:00Z"), "grade" : "A", "score" : 10 },
{ "date" : ISODate("2013-05-07T00:00:00Z"), "grade" : "A", "score" : 13 },
{ "date" : ISODate("2012-09-11T00:
00:00Z"), "grade" : "B", "score" : 18 }, { "date" : ISODate("2012-04-
18T00:00:00Z"), "grade" : "A", "score" : 9 }, { "date" : ISODate("2011-09-
27T00:00:00Z"), "grade" : "A", "score" : 8 }, { "date" : ISODate("2011-05-
03T00:00:00Z"), "grade" : "C", "score" : 77 } ], "name" : "Fortunato Bros Cafe &
Bakery", "restaurant_id" : "40400561" }

```

```
{ "_id" : ObjectId("564c2d949eb21ad392f180b0"), "address" : { "building" : "231",
"coord" : [ -73.9772294, 40.7527262 ], "street" : "Grand Central Station", "zipcode" :
"10017" }, "borough" : "Manhattan", "cuisine" : "Italian", "grades" : [ { "date" :
ISODate("2015-01-07T00:00:00Z"), "grade" : "Z", "score" : 20 }, { "date" :
ISODate("2014-07-03T00:00:00Z"), "grade" : "B", "score" : 14 }, { "date" :
ISODate("2013-12-21T00:00:00Z"), "grade" : "A", "score" : 13 }, { "date" :
ISODate("2013-05-17T00:
00:00Z"), "grade" : "C", "score" : 76 }, { "date" : ISODate("2012-04-
20T00:00:00Z"), "grade" : "A", "score" : 12 } ], "name" : "Two Boots Grand
Central", "restaurant_id" : "40725591" }

{ "_id" : ObjectId("564c2d949eb21ad392f186d3"), "address" : { "building" : "7705",
"coord" : [ -73.8878704, 40.7435875 ], "street" : "Woodside Avenue", "zipcode" :
"11373" }, "borough" : "Queens", "cuisine" : "Thai", "grades" : [ { "date" :
ISODate("2014-05-12T00:00:00Z"), "grade" : "B", "score" : 14 }, { "date" :
ISODate("2013-11-20T00:00:00Z"), "grade" : "C", "score" : 84 }, { "date" :
ISODate("2012-09-18T00:00:00Z"), "grade" : "A", "score" : 11 }, { "date" :
ISODate("2012-05-03T00:00:00Z"), "
grade" : "B", "score" : 23 } ], "name" : "Spicy Shallot", "restaurant_id" :
"40979431" }

{ "_id" : ObjectId("564c2d949eb21ad392f1890c"), "address" : { "building" : "566",
"coord" : [ -73.987977, 40.755195000000001 ], "street" : "7 Avenue", "zipcode" :
"10018" }, "borough" : "Manhattan", "cuisine" : "Chinese", "grades" : [ { "date" :
ISODate("2014-09-23T00:00:00Z"), "grade" : "A", "score" : 7 }, { "date" :
ISODate("2013-11-20T00:00:00Z"), "grade" : "B", "score" : 23 }, { "date" :
ISODate("2013-04-29T00:00:00Z"), "grade" : "C", "score" : 77 }, { "date" :
ISODate("2012-10-23T00:00:00Z")
, "grade" : "B", "score" : 21 }, { "date" : ISODate("2012-05-14T00:00:00Z"),
"grade" : "B", "score" : 21 }, { "date" : ISODate("2011-10-26T00:00:00Z"), "grade" :
"B", "score" : 23 } ], "name" : "Midtown Buffet", "restaurant_id" : "41038679" }

{ "_id" : ObjectId("564c2d949eb21ad392f18c36"), "address" : { "building" : "181",
"coord" : [ -73.977573999999999, 40.67127 ], "street" : "7 Avenue", "zipcode" :
```

```

"11215" }, "borough" : "Brooklyn", "cuisine" : "Pizza", "grades" : [ { "date" :
ISODate("2014-09-23T00:00:00Z"), "grade" : "A", "score" : 10 }, { "date" :
ISODate("2014-04-16T00:00:00Z"), "grade" : "B", "score" : 27 }, { "date" :
ISODate("2013-08-22T00:00:00Z"), "grade" : "A", "score" : 12 }, { "date" :
ISODate("2012-12-27T00:00:00Z"), "
grade" : "B", "score" : 22 }, { "date" : ISODate("2012-06-22T00:00:00Z"), "grade" :
"C", "score" : 77 }, { "date" : ISODate("2011-11-22T00:00:00Z"), "grade" : "A",
"score" : 12 } ], "name" : "Pino'S La Forchetta", "restaurant_id" : "41129343" }
{ "_id" : ObjectId("564c2d949eb21ad392f18da1"), "address" : { "building" : "499",
"coord" : [ -73.9651337, 40.6937213 ], "street" : "Myrtle Avenue", "zipcode" :
"11205" }, "borough" : "Brooklyn", "cuisine" : "Mexican", "grades" : [ { "date" :
ISODate("2015-01-14T00:00:00Z"), "grade" : "Z", "score" : 17 }, { "date" :
ISODate("2014-05-08T00:00:00Z"), "grade" : "C", "score" : 73 }, { "date" :
ISODate("2013-04-01T00:00:00Z"), "grade" : "A", "score" : 10 }, { "date" :
ISODate("2012-08-24T00:00:00Z"),
"grade" : "B", "score" : 25 }, { "date" : ISODate("2012-02-29T00:00:00Z"),
"grade" : "A", "score" : 8 } ], "name" : "Los Pollitos Iii", "restaurant_id" :
"41163308" }
{ "_id" : ObjectId("564c2d949eb21ad392f19172"), "address" : { "building" : "366",
"coord" : [ -73.981843, 40.741207 ], "street" : "3 Avenue", "zipcode" : "10016" },
"borough" : "Manhattan", "cuisine" : "Japanese", "grades" : [ { "date" :
ISODate("2014-02-27T00:00:00Z"), "grade" : "A", "score" : 10 }, { "date" :
ISODate("2013-09-10T00:00:00Z"), "grade" : "A", "score" : 10 }, { "date" :
ISODate("2013-03-08T00:00:00Z"), "grade" : "A", "score" : 13 }, { "date" :
ISODate("2012-08-13T00:00:00Z"), "gra
de" : "B", "score" : 26 }, { "date" : ISODate("2012-02-02T00:00:00Z"), "grade" :
"B", "score" : 18 }, { "date" : ISODate("2011-09-28T00:00:00Z"), "grade" : "C",
"score" : 76 }, { "date" : ISODate("2011-05-12T00:00:00Z"), "grade" : "A", "score" :
12 } ], "name" : "East Japanese Restaurant", "restaurant_id" : "41242254" }
{ "_id" : ObjectId("564c2d949eb21ad392f19400"), "address" : { "building" : "104-
11", "coord" : [ -73.861317, 40.7502198 ], "street" : "Roosevelt Avenue", "zipcode" :

```

```
"11368" }, "borough" : "Queens", "cuisine" : "Latin (Cuban, Dominican, Puerto
Rican, South & Central American)", "grades" : [ { "date" : ISODate("2014-08-
22T00:00:00Z"), "grade" : "A", "score" : 9 }, { "date" : ISODate("2014-03-
10T00:00:00Z"), "grade" : "A", "score" : 9 }, { "date" : ISODate("2013-09-
26T00:00:00Z"), "grade" : "A", "
score" : 11 }, { "date" : ISODate("2013-03-20T00:00:00Z"), "grade" : "C", "score" :
73 }, { "date" : ISODate("2012-09-24T00:00:00Z"), "grade" : "A", "score" : 10 },
{ "date" : ISODate("2012-04-18T00:00:00Z"), "grade" : "B", "score" : 21 } ],
"name" : "Los Mismo Restaurant", "restaurant_id" : "41288811" }
{ "_id" : ObjectId("564c2d949eb21ad392f1948d"), "address" : { "building" : "3845",
"coord" : [ -73.94352959999999, 40.8361392 ], "street" : "Broadway", "zipcode" :
"10032" }, "borough" : "Manhattan", "cuisine" : "Seafood", "grades" : [ { "date" :
ISODate("2014-09-18T00:00:00Z"), "grade" : "Z", "score" : 11 }, { "date" :
ISODate("2014-01-10T00:00:00Z"), "grade" : "B", "score" : 27 }, { "date" :
ISODate("2013-07-05T00:00:00Z"), "grade" : "C", "score" : 5 }, { "date" :
ISODate("2013-01-16T00:00:00Z"), "grade" : "A", "score" : 12 }, { "date" : ISODate("2012-07-17T00:00:00Z"),
"grade" : "B", "score" : 26 }, { "date" : ISODate("2012-03-14T00:00:00Z"), "grade" :
"C", "score" : 77 } ], "name" : "Brothers Fish Market", "restaurant_id" : "41297769" }
{ "_id" : ObjectId("564c2d949eb21ad392f19c8b"), "address" : { "building" : "222",
"coord" : [ -73.9579033, 40.7332745 ], "street" : "Franklin Street", "zipcode" :
"11222" }, "borough" : "Brooklyn", "cuisine" : "Italian", "grades" : [ { "date" :
ISODate("2014-12-11T00:00:00Z"), "grade" : "Z", "score" : 18 }, { "date" :
ISODate("2013-10-31T00:00:00Z"), "grade" : "A", "score" : 12 }, { "date" :
ISODate("2013-03-20T00:00:00Z"), "grade" : "B", "score" : 24 }, { "date" :
ISODate("2012-07-18T00:00:00Z"), "grade" : "B", "score" : 18 }, { "date" : ISODate("2012-01-12T00:00:00Z"),
"grade" : "C", "score" : 81 } ], "name" : "Anella", "restaurant_id" : "41410058" }
{ "_id" : ObjectId("564c2d949eb21ad392f1a8b9"), "address" : { "building" : "29",
"coord" : [ -73.9087571, 40.8539355 ], "street" : "West Burnside Avenue",
"zipcode" : "10453" }, "borough" : "Bronx", "cuisine" : "Chinese", "grades" :
```

```
[ { "date" : ISODate("2014-10-21T00:00:00Z"), "grade" : "B", "score" : 15 },
  { "date" : ISODate("2014-03-05T00:00:00Z"), "grade" : "A", "score" : 7 }, { "date" :
ISODate("2013-07-24T00:00:00Z"), "grade" : "A", "score" : 6 }, { "date" :
ISODate("2013-02-12T00:00:00Z")
, "grade" : "A", "score" : 12 }, { "date" : ISODate("2012-07-24T00:00:00Z"),
"grade" : "C", "score" : 73 }, { "date" : ISODate("2012-02-28T00:00:00Z"), "grade" :
"B", "score" : 23 } ], "name" : "Ming Wong Restaurant", "restaurant_id" :
"41563476" }
{ "_id" : ObjectId("564c2d949eb21ad392f1a951"), "address" : { "building" : "1724",
"coord" : [ -73.94981, 40.780043 ], "street" : "2 Avenue", "zipcode" : "10128" },
"borough" : "Manhattan", "cuisine" : "Indian", "grades" : [ { "date" :
ISODate("2014-09-25T00:00:00Z"), "grade" : "A", "score" : 7 }, { "date" :
ISODate("2014-03-20T00:00:00Z"), "grade" : "A", "score" : 12 }, { "date" :
ISODate("2013-09-09T00:00:00Z"), "grade" : "B", "score" : 21 }, { "date" :
ISODate("2013-03-25T00:00:00Z"), "grade"
: "B", "score" : 18 }, { "date" : ISODate("2012-08-15T00:00:00Z"), "grade" : "A",
"score" : 11 }, { "date" : ISODate("2011-12-23T00:00:00Z"), "grade" : "C", "score" :
98 } ], "name" : "Baluchi'S Indian Food", "restaurant_id" : "41569277" }
{ "_id" : ObjectId("564c2d949eb21ad392f1abd8"), "address" : { "building" : "844",
"coord" : [ -73.970671, 40.7515735 ], "street" : "2 Avenue", "zipcode" : "10017" },
"borough" : "Manhattan", "cuisine" : "Pizza", "grades" : [ { "date" : ISODate("2014-
07-15T00:00:00Z"), "grade" : "A", "score" : 12 }, { "date" : ISODate("2013-07-
15T00:00:00Z"), "grade" : "A", "score" : 10 }, { "date" : ISODate("2013-02-
19T00:00:00Z"), "grade" : "A", "score" : 13 }, { "date" : ISODate("2012-08-
02T00:00:00Z"), "grade
" : "A", "score" : 11 }, { "date" : ISODate("2012-02-29T00:00:00Z"), "grade" : "B",
"score" : 14 }, { "date" : ISODate("2011-09-02T00:00:00Z"), "grade" : "C", "score" :
79 } ], "name" : "La Trattoria", "restaurant_id" : "41595287" }
{ "_id" : ObjectId("564c2d949eb21ad392f1ad35"), "address" : { "building" : "194-
03", "coord" : [ -73.7874367, 40.7581296 ], "street" : "Northern Boulevard",
"zipcode" : "11358" }, "borough" : "Queens", "cuisine" : "Korean", "grades" :
```

```
[ { "date" : ISODate("2014-09-16T00:00:00Z"), "grade" : "C", "score" : 78 },
  { "date" : ISODate("2014-01-09T00:00:00Z"), "grade" : "B", "score" : 23 }, { "date" :
ISODate("2013-05-22T00:00:00Z"), "grade" : "A", "score" : 9 }, { "date" :
ISODate("2012-11-27T00:00:00
Z"), "grade" : "A", "score" : 12 }, { "date" : ISODate("2012-05-16T00:00:00Z"),
"grade" : "B", "score" : 27 } ], "name" : "Gal Bi Ma Eul", "restaurant_id" :
"41608854" }
{ "_id" : ObjectId("564c2d949eb21ad392f1adde"), "address" : { "building" : "56",
"coord" : [ -73.99836309999999, 40.7158265 ], "street" : "Mott Street", "zipcode" :
"10013" }, "borough" : "Manhattan", "cuisine" : "Chinese", "grades" : [ { "date" :
ISODate("2014-11-18T00:00:00Z"), "grade" : "Z", "score" : 25 }, { "date" :
ISODate("2014-03-31T00:00:00Z"), "grade" : "B", "score" : 22 }, { "date" :
ISODate("2013-07-12T00:00:00Z"), "grade" : "A", "score" : 9 }, { "date" :
ISODate("2013-02-08T00:00:00
Z"), "grade" : "C", "score" : 73 }, { "date" : ISODate("2011-12-07T00:00:00Z"),
"grade" : "A", "score" : 10 } ], "name" : "Wonton Noodle Garden", "restaurant_id" :
"41614664" }
{ "_id" : ObjectId("564c2d949eb21ad392f1b47e"), "address" : { "building" : "1191",
"coord" : [ -73.953931, 40.68096 ], "street" : "Fulton Street", "zipcode" : "11216" },
"borough" : "Brooklyn", "cuisine" : "African", "grades" : [ { "date" :
ISODate("2015-01-07T00:00:00Z"), "grade" : "Z", "score" : 38 }, { "date" :
ISODate("2014-06-26T00:00:00Z"), "grade" : "A", "score" : 9 }, { "date" :
ISODate("2013-12-26T00:00:00Z"), "grade" : "A", "score" : 9 }, { "date" :
ISODate("2013-12-02T00:00:00Z"), "gr
ade" : "P", "score" : 4 }, { "date" : ISODate("2013-06-13T00:00:00Z"), "grade" : "B",
"score" : 16 }, { "date" : ISODate("2013-01-03T00:00:00Z"), "grade" : "C", "score" :
58 }, { "date" : ISODate("2012-05-24T00:00:00Z"), "grade" : "C", "score" : 78 } ],
"name" : "Cheikh Umar Futiyu Restaurant", "restaurant_id" : "41660581" }
{ "_id" : ObjectId("564c2d949eb21ad392f1b688"), "address" : { "building" : "43-
46", "coord" : [ -73.9224579, 40.7441205 ], "street" : "42 Street", "zipcode" :
"11104" }, "borough" : "Queens", "cuisine" : "Japanese", "grades" : [ { "date" :
```

```
ISODate("2014-07-31T00:00:00Z"), "grade" : "B", "score" : 22 }, { "date" :
ISODate("2014-01-02T00:00:00Z"), "grade" : "A", "score" : 12 }, { "date" :
ISODate("2013-06-05T00:00:00Z"), "grade" : "B", "score" : 14 }, { "date" :
ISODate("2012-09-14T00:00:00Z"), "g
rade" : "C", "score" : 73 } ], "name" : "Takesushi", "restaurant_id" : "41679611" }
```

- 18. Write a MongoDB query to find the restaurant Id, name, address and geographical location for those restaurants where 2nd element of coord array contains a value which is more than 42 and upto 52.**

Structure of 'restaurants' collection :

```
{
  "address": {
    "building": "1007",
    "coord": [ -73.856077, 40.848447 ],
    "street": "Morris Park Ave",
    "zipcode": "10462"
  },
  "borough": "Bronx",
  "cuisine": "Bakery",
  "grades": [
    { "date": { "$date": 1393804800000 }, "grade": "A", "score": 2 },
    { "date": { "$date": 1378857600000 }, "grade": "A", "score": 6 },
    { "date": { "$date": 1358985600000 }, "grade": "A", "score": 10 },
    { "date": { "$date": 1322006400000 }, "grade": "A", "score": 9 },
    { "date": { "$date": 1299715200000 }, "grade": "B", "score": 14 }
  ],
  "name": "Morris Park Bake Shop",
  "restaurant_id": "30075445"
}
```

Query:

```
db.restaurants.find(
    {
        "address.coord.1": {$gt : 42, $lte : 52}
    },
    {"restaurant_id" : 1,"name":1,"address":1,"coord":1}
);
```

Output:

```
{ "_id" : ObjectId("564c2d939eb21ad392f1786b"), "address" : { "building" : "47",
"coord" : [ -78.877224, 42.89546199999999 ], "street" : "Broadway @ Trinity Pl",
"zipcode" : "10006" }, "name" : "T.G.I. Friday'S", "restaurant_id" : "40387990" }
{ "_id" : ObjectId("564c2d939eb21ad392f178c7"), "address" : { "building" : "1",
"coord" : [ -0.7119979, 51.6514664 ], "street" : "Pennplaza E, Penn Sta", "zipcode" :
"10001" }, "name" : "T.G.I. Fridays", "restaurant_id" : "40388936" }
{ "_id" : ObjectId("564c2d949eb21ad392f17af2"), "address" : { "building" : "3000",
"coord" : [ -87.86567699999999, 42.61150920000001 ], "street" : "47 Avenue",
"zipcode" : "11101" }, "name" : "Di Luvio'S Deli", "restaurant_id" : "40402284" }
{ "_id" : ObjectId("564c2d949eb21ad392f17d28"), "address" : { "building" :
"21972199", "coord" : [ -78.589606, 42.8912372 ], "street" : "Broadway", "zipcode" :
"10024" }, "name" : "La Caridad 78", "restaurant_id" : "40568285" }
{ "_id" : ObjectId("564c2d949eb21ad392f183f6"), "address" : { "building" : "7981",
"coord" : [ -84.9751215, 45.4713351 ], "street" : "Hoyt Street", "zipcode" :
"11201" }, "name" : "Bijan'S", "restaurant_id" : "40876618" }
{ "_id" : ObjectId("564c2d949eb21ad392f1840c"), "address" : { "building" : "0",
"coord" : [ -88.0778799, 42.4154769 ], "street" : "& Grand Central", "zipcode" :
"10017" }, "name" : "Hyatt, Ny Central/Room Service", "restaurant_id" :
"40879243" }
```

```
{ "_id" : ObjectId("564c2d949eb21ad392f18420"), "address" : { "building" : "60",
"coord" : [ -111.9975205, 42.0970258 ], "street" : "West Side Highway", "zipcode" :
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"11371" }, "name" : "Dunkin' Donuts", "restaurant_id" : "41142675" }
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"10018" }, "name" : "Chipotle Mexican Grill", "restaurant_id" : "41223834" }
{ "_id" : ObjectId("564c2d949eb21ad392f1909b"), "address" : { "building" : "350",
"coord" : [ -93.2069217, 43.14769159999999 ], "street" : "5 Avenue", "zipcode" :
"10018" }, "name" : "Europa Cafe", "restaurant_id" : "41229799" }
{ "_id" : ObjectId("564c2d949eb21ad392f192d9"), "address" : { "building" : "350",
"coord" : [ -93.2069217, 43.14769159999999 ], "street" : "5 Avenue", "zipcode" :
"10018" }, "name" : "Ny Skyride", "restaurant_id" : "41266025" }
{ "_id" : ObjectId("564c2d949eb21ad392f194a6"), "address" : { "building" : "1",
"coord" : [ -72.4751457, 43.2956803 ], "street" : "Wall Street Court", "zipcode" :
"10005" }, "name" : "Haru", "restaurant_id" : "41298810" }
{ "_id" : ObjectId("564c2d949eb21ad392f195a7"), "address" : { "building" : "1618",
"coord" : [ -77.6873462, 43.2236789 ], "street" : "Maiden Lane", "zipcode" :
"10038" }, "name" : "Oka Sushi", "restaurant_id" : "41312239" }
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"coord" : [ -93.2069217, 43.14769159999999 ], "street" : "5 Avenue", "zipcode" :  
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"zipcode" : "10018" }, "name" : "Pier Side Cafe", "restaurant_id" : "41367417" }  
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"11103" }, "name" : "Mexi Q Kitchen", "restaurant_id" : "41515837" }
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# LIVEWIRE